Ajay G

+91 9952053743
ajayg@iisc.ac.in

S3- Victory Flat, 34/35-Perumal Koil Street,
West Mambalam, Chennai-600033, INDIA

Skills

Programming	**C/C++, Python**, MATLAB
Framework	**ROS**, **POSIX**, Orocos
OS	**Ubuntu**, Windows, **ChibiOS (Embedded RTOS)**
Simulation	**MoveIt, Gazebo,** ANSYS Workbench
Version & Build	Git, GitHub, Make, CMake, Catkin
Debugging Tools	**OpenOCD, gdb**, Valgrind, strace, ptrace
Single Board Computer	Raspberry-Pi, Odroid, **Beaglebone**
Microcontroller	**STM32F4 series** (ARM Cortex-M4), Arduino
Bus & Network	CAN, I2C, SPI, RS-232, Socket(TCP/IP), MQTT
Others	**Doxygen**, LaTeX, MS-Office, Prezi, Solidworks

Areas of Interest

Software development for Real Time Motion Planning and Control of Bio-inspired & Underactuated Robots

Software Experience

1. Indian Institute of Science, Bangalore — Feb 2018-present
   Project Assistant — Robert Bosch Centre for Cyber-Physical System

   *I'm part of the team which is developing **quadruped robot** (a hybrid system). My role is to develop a custom Real Time Firmware for position, and torque, control of* **BLDC *drive*** *and to implement the planning algorithm in Real Time on Odroid using Orocos Framework and integrate it with the* **CAN *bus***.

2. ASIMOV Robotics Pvt. Ltd, Kochi — Jul 2017 to Dec 2017
   Internship — Software Development (R&D Division)

   *We developed a **banking service robot**, and **arm manipulators**, for specific clients. My role was to interface actuators* (**Dynamixel smart servo**, DC motor) *and sensors* (lidar, IMU) *with* **ROS** *and to perform motion planning both on real hardware and in simulation in* **Gazebo***, and* **MoveIt**.

Design Experience

1. TIDC INDIA (A Unit of Tube Investments of India Ltd.), Ambattur — July 2016
   Internship
   Design of simple cam chain in Solidworks and did dynamic motion analysis of it in ANSYS Workbench

2. J.K. Fenner (India)Ltd, sriperumbudur — June 2016
   Internship
   Design of 100-ton Hydraulic Press in Solidworks and performed both Static Load, and Thermal, analysis in ANSYS Workbench.

Manufacturing Experience

1. Ashok Leyland, Ennore — June 2015
   In-plant Training
   Had a practical exposure to various manufacturing methods and assemble line production system.

Major Achievement

Finalist (Top 10 team out of 11,000+ applicants) in DST & Texas Instruments Inc., India Innovation Challenge Design Contest 2016 anchored by IIM Bangalore for our project titled, "Smart Intravenous Dripper"

The goal was to prevent the back flow of blood and air embolism/blockage caused due to drainage of Intravenous drippers bags by using a capacitive sensor to monitor the liquid level and if the Intravenous bag runs out of the glucose then immediately the cam actuated by servo stops the flow thus preventing any accident to the patient and will notify the nurse for further replacement of the IV bag.

The project was made as a Startup and was **funded (20,00,000 INR) by DST, Govt of India** and is currently incubated at IIM Bangalore as Rekindle Automation Pvt. Ltd.

Education

B.Tech in Mechanical Engineering	B.S. Abdur Rahman University	2013-2017	8.45/10.0
12th Grade, TN State Board	GRT, Chennai	Mar 2013	88.0 %
10th Grade, TN Matriculation Board	AVM, Chennai	Mar 2011	87.4 %

Honors & Awards Leadership

- ISHRAE 2016-17 President in B.S. Abdur Rahman University
- Won 1st Prize in ISHRAE Chennai chapter Level Quiz by St. Joseph's College of Engineering, Chennai under the auspices of ISHRAE
- Won 1st Prize in Technical Quiz at Confluence 2016, A National Level Technical Symposium
- Selected as Finalist in the Semifinals of AQuest, An All India Inter-Collegiate Engineering Quiz Contest at Kolkata by ISHRAE
- Won 1st Prize in Design of Heat Exchanger at Mechfest 2016, Technical Symposium
- Participated in Finals of AQuest, An All India Inter-Collegiate Engineering Quiz Contest at Mumbai by ISHRAE
- Won 1st Prize in Tamil Oration at A.V. Meiyappan Matriculation Higher Secondary School
- Successfully Passed- Business English Certificate-BEC Preliminary, Council of Europe Level B1 by Cambridge English Language Assessment. (Part of the University of Cambridge)

Extra-Curricular Achievement

- Won 3rd Prize in Choreography at Festeve 2014 by Women's Christian College, Chennai
- Won 3rd Prize in Burn the floor at Chrysalis 2014 by Loyola Institute of Business Administration, Chennai
- Won 3rd Prize in Western Dance at Srishti 2014 by Ethiraj Collage for Women, Chennai

References

Dr. Shishir N. Y. Kolathaya
INSPIRE Faculty fellow,
Robert Bosch Center for Cyber-Physical Systems
Third Floor, SID Building, Indian Institute of Science,
Bengaluru, Karnataka 560012
E-mail id: shishirk@iisc.ac.in

Mr. Jayakrishnan T
Chief Executing Officer,
ASIMOV Robotics Pvt. Ltd,
1st Floor, KTIZ, Kinfra-Hi Tech Park,
Kalamassery, Kochi, Kerala 683503
Email id: jk@asimovrobotics.com